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ABOUT ME

Software Engineer with over **7 years** of hands-on expertise in **Fullstack development along with AI solutions**. Proven track record of building scalable, secure, and high-performance systems, integrating complex pipelines, and delivering innovative software that drives real-world impact. Adaptable, quick to learn new technologies, and committed to pushing the boundaries of software development to create cutting-edge solutions.

EDUCATION AND TRAINING

BS Software Engineering

Punjab University College of Information and Technology [2015 – 2019]

City: Lahore | Country: Pakistan | Field(s) of study: Information and Communication Technologies | Final grade: 3.58

WORK EXPERIENCE

🏢 *Valohai Oy – Finland*

Senior Software Engineer (Django, React, AI)

[05/2023 – 12/2025]

- Served as a key fullstack developer on a **high-impact MLOps platform** -> app.valohai.com with **React Typescript** frontend and **Python Django** backend
- Handling complex **pipelines, tasks, multiple cloud integrations** and **docker containers**
- Responsible for designing and optimizing APIs, developing robust frontend components, integrating systems seamlessly, and delivering **efficient, secure, and scalable solutions** that drive real-world client usage.
- Moreover worked on **sample multiple AI/ML projects** used for demo/trial on Valohai platform
- Contributed to Valohai's **open-source** repositories
- Every month for a week was responsible for solving real-time customer questions/reported bugs and feedback.

🏢 *Storewide Active Intelligence – United Kingdom*

Machine Learning Engineer (Flask, FastAPI, React, Docker)

[09/2020 – 04/2023]

- Responsible for fullstack development of modules (Frontend, backend, AI layer, computer vision algorithms)
- Designed and implemented a **theft prevention system** for retail aisles, enabling real-time insights into customer activity and item interaction in multiple aisles. The system efficiently handled **up to 8 camera feeds per CPU**.
- Developed a **stock alert system** that automatically notifies store staff when inventory levels are **low or empty**, improving product availability and shelf management.
- Built a **real-time theft detection solution** integrated with **self-checkout systems** syncing the POS data with camera feed, reducing shrinkage through AI-based behavior monitoring.
- Worked as **project owner** for multiple computer vision and AI-based solutions, delivering end-to-end implementations on a **weekly release cycle**
- Worked closely with stakeholders for handling on-premises deployment constraints

🏢 *Xavor Corporation – Pakistan*

Software Engineer (Python, C#, React, SPFx)

[06/2019 – 10/2020]

- Worked as a fullstack developer, building end-to-end solutions by developing a unified search platform and internal IoT/sales management portal using **React-Redux, Python, Django REST, Azure Search, Azure Table Storage, and Azure CosmosDB**.
- Automated enterprise processes, optimized application performance, delivered **data analytics and reporting**, and ensured best practices throughout development.

- Developed custom **SharePoint solutions using SPFx and React**, including reusable web parts and UI components tailored to business requirements. Worked on integrating modern frontend practices within the SharePoint ecosystem to enhance usability and internal workflows.
- Was part of 2 teams throughout the period, the **consultation services team** and the **product team**.

Depository Works – United States

Backend Developer

[09/2018 – 05/2019]

My responsibilities included development, debugging, management and deployment of sales website of company. Additionally, I was responsible for 3rd party API integrations (Jet, Wish, AliBaba, Amazon) into shopify.

AIM Research Lab

Internee

[08/2018 – 05/2019]

Being a part of lab, I worked in a team and we followed SCRUM throughout the development of our final year project.

Punjab University College of Information Technology

Teacher Assistant

I was appointed as teaching assistant for the courses Object Oriented Programming and Programming Fundamentals. My responsibilities included conducting labs, preparing and grading the quizzes and assignments. In addition to that, I was also responsible for assisting students in my free time.

PROJECTS

Valohai Libraries Integration

- Led the end to end development of a **reusable library integration** feature across frontend, backend, and APIs.
- Enabled users to **integrate external repositories** as shared Valohai libraries without onboarding them as primary projects.
- Built **GitHub integration** for seamless repository discovery, linking, and synchronization.
- Designed and implemented role based **access control and permission management**.
- Developed secure library ownership and access handling across teams and projects.
- Enabled library steps to run as **isolated containers**, ensuring scalability and execution independence.
- Improved reusability of **ML workflow** components and reduced duplication across repositories.
- Collaborated across the stack to deliver a production ready feature used by multiple teams.

Theft Alert System

1. We trained a **spatio-temporal activity recognition model** using **transfer learning** on **custom suspicious activities** (putting item in pocket, hiding something in jacket or personal bag etc) data collected from **real-world surveillance cameras**. It was trained over 450+ theft activity videos
2. Built a config file for custom training of **SlowFast** model and prepared the structured data in supervised manner
3. Deployed the model on system using **multi threading** to increase the speed, It works on real-time camera feed and notifies security staff with the concerned person's short video if any suspicious activity is recognized in any part of the store

Link: <https://saigroups.co.uk/#aisle>

Vision based Customer Analytics

1. The purpose of this project was to analyze customer behavior in different aisles
2. Built a module which **detects and track a person** in a video stream, analyzing whether he is shopping or just passing by the aisle
3. **Custom trained Classification model** is being used to distinguish between passers by, shoppers and store staff
4. On each interaction with aisle system **smartly** generates a report containing following information
 - How much time a person spent in aisle (dwell time)
 - How many interactions he made with items in aisles (aisle engagement)
 - How many persons just passed by the aisle without looking
 - How many person actually stayed to check the stock

Stock Alert Generator

1. Built a alert generation system if stock in particular aisle is either empty or low
2. Used different computer vision techniques (edge detection, contour detections etc) to analyze whether any space in aisle is entirely empty or low
3. Developed a cron job to run the script every 30 minutes to generate emails/notifications on mobile app if some items are out of stock

Mesh Drive (Final Year Project)

1. Developed a single storage drive simulation by integrating four giants storage drives: Google Drive, Dropbox, One-Drive and SoundCloud
2. User could integrate **multiple accounts** for above mentioned all drives and then all of his data could be seen on **single platform**
3. We provided multiple smart option to do **filtering, sorting and searching** of data
4. I was mainly responsible for the **Smart Image search module**.
5. Image search via **k-mean clustering** was added. Moreover, **object detection (Yolov5)** and **custom trained face recognizer** model were also added to enhance the vision based search
6. **SCRUM framework** was followed throughout the development of the project
7. The project was done with the collaboration of [Artificial Intelligence and Multidisciplinary \(AIM\) Research Laboratory](#)

HONOURS AND AWARDS

Silver Medalist I received Silver Medal in my matriculation exams for achieving 2nd position in overall district in Secondary SchoolCertificate Exams, Lahore.

Clash of Coders, Speed Programming Competition, SoftExpo'17 I and my team secured 1st rank based on accuracy of results and speed in a national level competition. Over 80 teams participated from all over the Pakistan.

ICPC Pakistan Multi-Provisional Programming Contest, Softcom'18 I was 1st runner up in a national level speed programming competition. It was held in one of the top universities of Pakistan, GIKI.

1st Runner Up in Catch a Bug , Debugging Competition Out of 50 participant teams, my team secured Rank 2

Winner, UCP Speed Programming Competition I won the Speed Programming Competition in 2018 held in University of Central Punjab with two more members of my team

Principal's Honorary List I was selected in the list and received a merit scholarship. It includes students who secure a semester GPA of 3.50/4.0 or higher. Mt GPA was 4.0/4.0

Visit to Youth Development Center (YDC) On securing [2nd position in matriculation](#) in overall district, government arranged a trip of all top position holders to [YDC Murree](#). There, I was included in many workshops for personal development.

Guest Speaker on National Television I was invited as guest speaker on International Youth day, as I was one of the [top position holders](#) in matriculation in 2013.

HOBBIES AND INTERESTS

Hobbies and interests ReadingSurfing Internet

CERTIFICATIONS

Convolutional Neural Networks

[\[Certificate\]](#)

Machine Learning by Andrew NG

[\[Certificate\]](#)

TECHNICAL SKILLS

Technical Skills

Languages: Python, C#, Java, SQL, Assembly, Matlab, Javascript, Typescript

Technologies & Frameworks: React, Flask, Django, Selenium, EC2(AWS), OpenCV, PyTorch, Keras, Redis, Celery, SPFx-Sharepoint Online, Node, MongoDB, .Net Core 2, .Net Core 3, FastAPI, AWS, Claude, Codex, Docker, Kubernetes, PostgreSQL, ExpressJS

Skills: ETL/ELT data pipelines, Git, Web Scrapping, SQL Databases, NoSQL Databases, Cloud functions, Azure Services, AI driven development, Containerization, Deep learning

MANAGEMENT AND LEADERSHIP SKILLS

Management and Leadership Skills

Publicity Assistant at ACM

I volunteered for a position as an [Assistant Publicity Head](#) of ACM Chapter PUCIT 2018. My responsibilities included involving as many students in our events as I could. I successfully formed a community of students to participate in our talks, workshops, and competitions. Moreover, whenever the event would be of national level, I with my team, used to share our events on all social platforms so that students from all over the country could engage.

General Secretary - Computer Society

I was appointed as General Secretary of Computer Society in 2018. In that tenure, I successfully organized a National Programming Competition in collaboration with ACM Chapter PUCIT in which more than 60 teams participated from different cities of Pakistan.

[Microsoft Insider Dev Tour](#) Organizer

I successfully organized a huge event of Microsoft with my team in 2019. The event has worldwide recognition. Worked under the supervision of [MVP](#) (Most Valuable Professional) of Microsoft. [[Verification](#)]